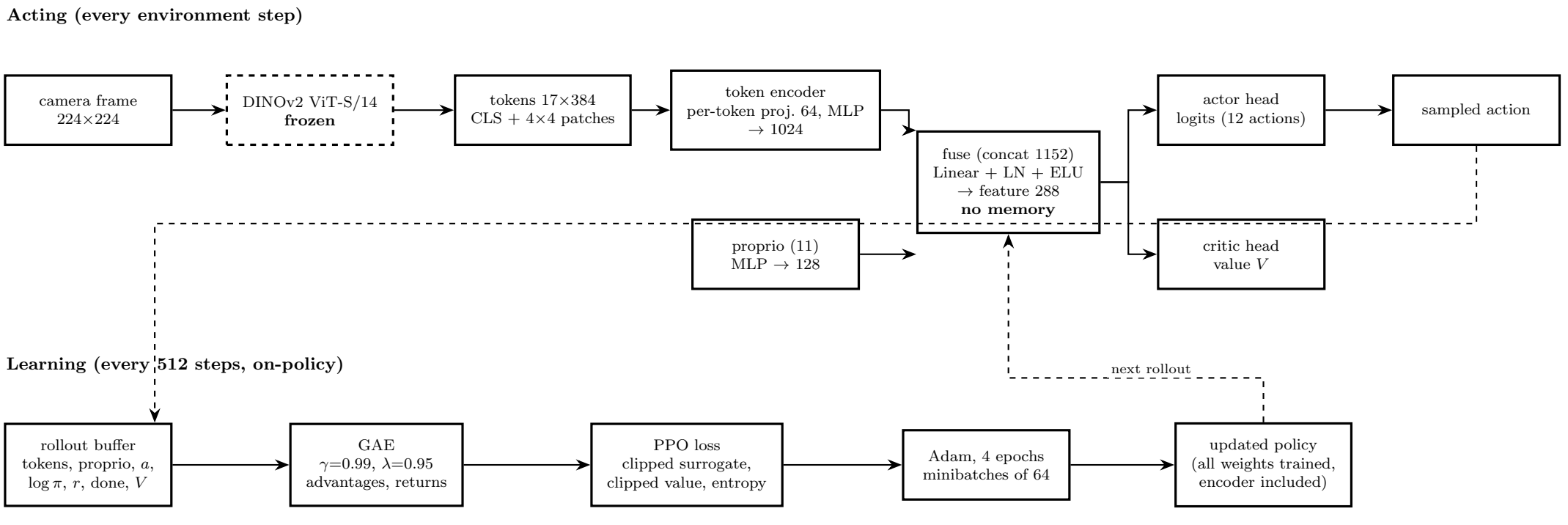

Acting (every environment step)
camera frame
224×224
DINOv2 ViT-S/14
frozen
tokens 17×384
CLS + 4×4 patches
token encoder
per-token proj. 64, MLP
→ 1024
fuse (concat 1152)
Linear + LN + ELU
→ feature 288
no memory
actor head
logits (12 actions)
sampled action
proprio (11)
MLP → 128
critic head
value V
next rollout
Learning (every 512 steps, on-policy)
rollout buffer
tokens, proprio, a,
log π, r, done, V
GAE
γ=0.99, λ=0.95
advantages, returns
PPO loss
clipped surrogate,
clipped value, entropy
Adam, 4 epochs
minibatches of 64
updated policy
(all weights trained,
encoder included)